Revision Diff Report: body.tex v1 $\rightarrow$ v2

Leverage Direction Matters

Date: 2026-03-30

Issues Addressed: H1 (Scope overload), H3 (In-sample threshold), H7 (Numerical inconsistencies)

Summary of Changes

Issue	Change Description
H1 (Scope overload)	Restructured from three contributions to two. TZ momentum removed from abstract, introduction, and conclusion as a contribution; retained as Appendix B with explicit framing as "supplementary evidence of information transmission."
H3 (In-sample threshold)	Added explicit in-sample vs. OOS caveat paragraph with uncertainty quantification. Reported OOS classification accuracy (6/6 and 5/6) separately from in-sample (9/9). Noted random baseline probability ($2^{-6} \approx 1.6\%$).
H7 (Numerical inconsistencies)	(a) Corrected "twelve DM comparisons" to "nine" throughout (matching Table 3's 9 rows). (b) Reconciled ρ values by attributing each to its specific sample ($\rho = 0.944$, $N = 5$ primary; $\rho = 0.83$, $N = 14$ extended). (c) Corrected "all four above / all eight below" to "all four above / all five below" (matching 9 total).

Detailed Changes

1. Abstract (main_v2.tex)

H7: ~~correctly classifying all twelve Diebold-Mariano comparisons across two out-of-sample periods~~ $\rightarrow$ **correctly classifying all nine Diebold-Mariano comparisons in the primary sample, with 6/6 correct out-of-sample predictions**

H7: ~~universal across all twelve tested assets and driven by base volatility level ($\rho = 0.944$)~~ $\rightarrow$ **universal across all tested assets and driven by base volatility level ($\rho = 0.944$ for primary assets, $\rho = 0.83$ for the extended $N = 14$ sample)**

H1: Removed all TZ momentum content from abstract. Added domain restriction clarification ($\rho = -0.448$, $p = 0.14$ for twelve assets).

2. Introduction (body_v2.tex, lines 9–17)

H1: ~~This paper makes three contributions~~ $\rightarrow$ **This paper makes two contributions**

H1: Entire "Third" contribution paragraph (TZ momentum, 8 lines) removed. Replaced with single sentence: **Supplementary evidence of cross-market information transmission is presented in Appendix B.**

H7: In First contribution: ~~all twelve Diebold-Mariano comparisons~~ $\rightarrow$ **all nine Diebold-Mariano comparisons across two out-of-sample periods (Table 3)**

H7: In Second contribution: Added explicit sample attribution for ρ values: **(Pearson $\rho = 0.944$, $p < 0.001$, $N = 5$ primary assets; Pearson $\rho = 0.83$, $p = 0.0002$, $N = 14$ in the extended sample)**

H1: Second contribution reframed as "application of the leverage taxonomy to VT" rather than an independent contribution.

3. Model Selection Rule (body_v2.tex, Section 4.3.2)

H7: ~~all twelve DM test outcomes across six assets and two OOS periods~~ $\rightarrow$ **all nine DM test outcomes across six assets and two OOS periods (Table 3)**

H7: ~~all four asset-period combinations above the threshold…while all eight combinations below~~ $\rightarrow$ **all four asset-period combinations above the threshold…while all five combinations below**

H7: ~~identical classifications for all twelve asset-period combinations~~ $\rightarrow$ **identical classifications for all nine asset-period combinations**

4. In-Sample Caveat (body_v2.tex, new paragraph after Section 4.3.2)

H3: Replaced brief one-sentence caveat with dedicated subsection "**In-sample versus out-of-sample classification.**" New content:

> **An important caveat is that both the numerical threshold ($\gamma > 0.08$) and the significance-based formulation ($t > 1.65$) are calibrated on the same sample used for evaluation; the 9/9 "perfect classification" is therefore partly in-sample and should not be interpreted as a measure of predictive accuracy. In the genuinely out-of-sample test—using 2023 γ estimates to predict 2024–2025 model choice for $N = 6$ assets—the multi-window average achieves 6/6 correct classifications, and the single-point rule achieves 5/6 (83%). However, with $N = 6$, even random classification would achieve 100% accuracy with probability $2^{-6} \approx 1.6\%$, so the OOS evidence, while supportive, has limited statistical power. Independent validation on a larger, separate asset universe is needed to confirm generalizability of both the threshold level and the classification accuracy.**

5. VT Cross-Asset Results (body_v2.tex, Section 4.4.2)

H7: ~~improvement strongly correlated with base volatility ($\rho = 0.83$, $p = 0.0002$, $N = 14$ assets…; Spearman $\rho = 0.88$)~~ $\rightarrow$ **For the five primary assets in Table 5, Pearson $\rho = 0.944$ ($p < 0.001$); for the extended sample of $N = 14$ assets (including eight additional commodities and international equities from Appendix A), Pearson $\rho = 0.83$ ($p = 0.0002$, Spearman $\rho = 0.88$).**

6. VT as Volatility Scaling (body_v2.tex, Section 4.4.5)

H7: ~~$\rho = 0.83$, $p = 0.0002$, $N = 14$~~ $\rightarrow$ **$\rho = 0.944$ ($p < 0.001$) for the five primary assets, $\rho = 0.83$ ($p = 0.0002$) in the extended $N = 14$ sample**

7. Conclusion (body_v2.tex, Section 6)

H1: ~~Three contributions emerge~~ $\rightarrow$ **Two contributions emerge**

H7: ~~correctly classifying all twelve Diebold-Mariano comparisons~~ $\rightarrow$ **correctly classifies all nine Diebold-Mariano comparisons in the primary sample (Table 3), with out-of-sample validation achieving 6/6 correct classifications…though the small cross-section ($N = 6$) limits the statistical power of this OOS test**

H1: Entire "Third" contribution paragraph (TZ momentum, 5 lines) replaced with single sentence: **Supplementary evidence on cross-market information transmission between U.S. and Asian equity markets is presented in Appendix B.**

H7: Reconciled ρ values: $\rho = 0.944$ **for** $N = 5$ **primary assets;** $\rho = 0.83$ **for the extended** $N = 14$ **sample**

H7: Added domain restriction note: **the mapping does not extend to diverse asset classes (**$\rho = -0.448$**,** $p = 0.14$ **for** $N = 12$**), delineating a clear domain restriction**

8. Appendix B Heading (body_v2.tex)

H1: ~~Time-Zone Information Transmission: US to Asian Markets~~ $\rightarrow$ **Time-Zone Information Transmission: Supplementary Evidence**

Added framing paragraph: **This appendix presents supplementary evidence on cross-market information transmission between U.S. and Asian equity markets. While not a main contribution of this paper, the lead-lag relationship provides independent evidence that the leverage-direction taxonomy has cross-market implications: U.S. equity volatility (driven by standard leverage) transmits to Asian markets with predictable timing.**

Files Modified

File	Status
`body.tex`	Unchanged (original preserved)
`body_v2.tex`	New file (revised body)
`main.tex`	Unchanged (original preserved)
`main_v2.tex`	New file (revised main, references body_v2)
`tables.tex`	Unchanged (no table data modified)
`table_nulls.tex`	Unchanged
`v1_to_v2_diff.tex`	This document

Remaining HIGH Issues (Not Addressed in This Revision)

H2. **Proposition 1 is partly mechanical** (G1) — requires either algebraic proof or independent volatility measure test

H4. **Overlapping windows inflate significance** (M2) — requires non-overlapping windows or block bootstrap

H5. **Yahoo Finance data source** (D1) — requires CRSP/Bloomberg switch

H6. **Short sample period** (D2) — requires unified extended sample

H8. **Zero-mean specification** (S1) — requires AR(1) robustness in main text

H9. **Proposition 1 reverses for diverse assets** (R3) — domain restriction now flagged more prominently (partial fix via H7)